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Enhancing Reservoir Characterization in Complex Laminated Formations: A Multiscale Evaluation of Wireline Formation Pressure Testing Using Mangahewa Field Case Study

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Abstract

Formation characterization is essential for maximizing well production and hydrocarbon recovery. It requires analyzing reservoir properties across multiple scales using tools with varying radii of investigation. In cases where traditional well testing methods like Drill Stem Testing (DST) are limited by cost, environmental concerns, or operational constraints, Wireline Formation Pressure Testing (WFPT) offers a viable alternative for near-wellbore evaluation. This case study assesses the reliability of reservoir parameters derived from pretest pressure data, using permeability as a proxy. The analysis benchmarks WFPT results against core data, log-based models, and DST across several zones within the laminated Mangahewa Formation in New Zealand's Taranaki Basin to evaluate interpretation accuracy in complex reservoirs.

This study focuses on the laminated, gas-bearing intervals of the Mangahewa Formation in the Taranaki Basin, where a spherical flow interpretation model was developed to extract both average permeability and anisotropy effects from WFPT pretest pressure responses. Permeability estimates were benchmarked against core plug measurements, NMR log interpretations, porosity-permeability transform models, and DST results. Six distinct reservoir intervals (MaD1-A, MaD1-B, MaC3, MaB1, MaA5u, MaA5l) in the Mangahewa-12 well were evaluated to assess the method's reliability in complex, heterogeneous formations.

WFPT-derived permeabilities ranged from sub-millidarcy to 60 mD and showed good agreement with core and NMR data in moderate-permeability zones. However, in tighter, anisotropic layers, WFPT diverged from log-based models, which tended to underestimate reservoir quality. DST results consistently reported lower permeabilities due to larger-scale averaging and completion skin effects. Pressure derivative analysis identified transitional flow regimes, including hemispherical and anisotropic behavior, though interpretation was occasionally constrained by short test durations and noise. These results demonstrate the quantitative value of WFPT for characterizing permeability and near-wellbore conditions in laminated reservoirs. The findings also emphasize the need for multiscale data integration and highlight key limitations and best practices for WFPT interpretation in complex geological settings.

This study delivers a detailed, zone-by-zone assessment of WFPT performance in a tight, laminated gas reservoir, highlighting how geological heterogeneity, flow geometry, and operational constraints influence permeability interpretation. By integrating analytical solutions, numerical simulations, log-based transforms, core data, and DST results across six distinct intervals, the analysis reinforces the value of a multi-method approach for reliable reservoir characterization in complex formations. The findings clarify both the strengths and limitations of WFPT in tight gas settings and offer practical guidance for its application in similarly challenging reservoirs.

Introduction

Reliable reservoir characterization is critical for effective hydrocarbon field development, particularly in tight and heterogeneous formations where small-scale permeability and pressure variations can significantly influence production strategy. Pressure transient analysis (PTA) and rate transient analysis (RTA), derived from tests such as drawdown, buildup, and drill stem testing (DST), remain indispensable for estimating permeability, skin, and diagnosing flow regimes (Elshahawi et al. 2007; Hashem et al. 2007). DSTs also play a critical role in assessing well deliverability and calculating the productivity index. However, their high cost, operational complexity, and environmental impact, particularly in offshore environments or early-stage appraisal wells, often limit their use (Jones et al. 2020; Ramaswami, Elshahawi, and El Battawy 2012). When separation facilities are available, extended DSTs can be conducted to obtain high-quality fluid samples and to define reservoir boundary conditions.

As an alternative to DST, wireline formation pressure testing (WFPT) offers a less invasive and more cost-effective method for evaluating exploration and appraisal wells, particularly in settings without surface separation facilities. Wireline tools such as the Modular Formation Dynamics Tester (MDT), whether deployed on wireline or drill-pipe, seal against the borehole wall, draw down formation fluids, and record the resulting pressure response during drawdown and buildup (Azari et al. 2016). A typical test sequence includes pad setting, a small pretest drawdown (typically 1–5 cm³), pressure buildup or falloff, and a larger main drawdown for fluid sampling. The pressure transient initially reflects spherical flow and may later transition to radial flow; in low-mobility formations, flow may remain hemispherical, with supercharging effects dominating. Accurate interpretation requires isolating the influence of mud cake resistance, tool storage, and formation mobility on the early-time slope. These tools also enable in-situ fluid analysis and compositional sampling. When performed at closely spaced depths, WFPT becomes a cornerstone of modern near-wellbore reservoir characterization in tight formations (Belhouchet et al. 2021).

Several studies have investigated the reliability of WFPT in estimating reservoir properties. (Mkhize, Aluko, and Gringarten 2016) found that WFPT-derived horizontal permeability can align with DST values after appropriate upscaling, although vertical permeability often requires harmonic averaging to account for spherical flow effects. Similarly, (Loi et al. 2012; Ramaswami, Elshahawi, and El Battawy 2012) reported that WFPT yields reasonable estimates in homogeneous formations but tends to underestimate permeability in laminated or low-permeability zones due to geometric distortion near the probe and pressure signal instability. (Motaei et al. 2023) emphasized that WFPT consistently underestimates permeability unless corrected using SCAL-derived adjustments. (Blinov et al. 2020) further noted challenges in fluid contact interpretation caused by short transients and incomplete fluid cleanup.

The short duration of pressure transients, combined with tool storage effects and measurement noise, often obscures diagnostic derivative plots and complicates flow regime identification (Ramaswami et al. 2016). To simulate non-radial flow behavior, modeling approaches such as Vertical Limited Entry (VLE) and Limited Entry (LE) have been proposed. However, even advanced numerical techniques such as the reverse-injection method introduced by (Jones et al. 2020) face limitations in representing heterogeneous pore structures and anisotropies unless supported by core, log, or DST data. Taken together, these studies

suggest that WFPT is most effective when used as part of a multi-method integration informed by geological context and high-resolution petrophysical data.

Despite previous efforts, a clear gap remains in demonstrating how WFPT can be reliably applied in laminated, gas-bearing systems, particularly in settings where DST is not feasible. This paper addresses that gap by evaluating the performance of WFPT in the Mangahewa Formation, a geologically complex gas reservoir in New Zealand's Taranaki Basin. The Mangahewa-12 well, which intersects multiple laminated intervals of the Eocene Mangahewa Formation, serves as the basis for a comparative evaluation across six gas-bearing zones. These intervals exhibit variable rock quality and strong vertical heterogeneity, making them representative of the formation's broader characteristics.

This study employs both analytical and numerical methods to interpret MDT pretest pressure data, incorporating spherical flow assumptions and limited entry flow modeling using commercial software. WFPT-derived permeability estimates are benchmarked against core plug measurements, NMR-based porosity–permeability transforms, and DST results. Key indicators such as the vertical permeability ratio (k_z/k_r) and hydraulic window (Motaei et al. 2023) are used to evaluate vertical communication and the influence of flow geometry.

The objective of this study is threefold: (1) to evaluate the extent to which WFPT can accurately characterize permeability in laminated gas reservoirs, (2) to identify the dominant sources of error and diagnostic uncertainty in pressure interpretation, and (3) to assess the value of integrating core, NMR, and DST data in constraining WFPT-derived permeability estimates. By quantifying both the strengths and limitations of WFPT under real field conditions, this work provides practical guidance for its use in similarly complex formations worldwide.

This study provides new insights into the diagnostic potential of WFPT in laminated and anisotropic gas reservoirs, offering a validated methodology for permeability estimation in settings where traditional well testing is limited. Early-time spherical or hemispherical flow often distorts pressure transients, leading to discrepancies between WFPT-derived permeability and values obtained from core analysis, DSTs, or log-based transforms. This issue is particularly critical in gas-bearing sequences with tight intervals and interbedded lithologies, where permeability can vary by orders of magnitude over vertical distances of less than one meter. Failure to resolve these variations accurately can lead to suboptimal well placement, inappropriate completions, and unreliable reservoir models. The findings contribute to more informed development decisions, optimized data acquisition strategies, and reduced uncertainty in early-phase reservoir modeling, delivering tangible value to industry and the broader geoscience community.

Methodology

Given the variety of methods available for estimating reservoir permeability (Ahmed, Crary, and Coates 1991), this study compares four approaches for characterizing both reservoir-scale and near-wellbore permeability, each employing different measurement mechanisms and interpretation techniques. These methods are categorized as either static or dynamic. Static methods include core plug measurements and log-based permeability interpretations, while dynamic methods involve pressure transient analysis derived from DST and WFPT data.

- a. **Core Data:** Porosity and permeability measurements from core plugs are a conventional part of core analysis, with standardized protocols well established (Warren, Skiba, and Price 1961). To ensure meaningful comparison with other methods, cores must be tested under reservoir conditions, as measurements at ambient pressure often overestimate permeability due to grain dilation and microfracture opening (Krause, Krevor, and Benson 2013). Steady-state or high-pressure gas tests are typically used for moderate- to high-permeability rocks, while tight formations require pulse-decay or pressure-transient methods. Multiple flow rates are used to confirm Darcy linearity, and Klinkenberg corrections are applied using gas data at varying mean pressures. Advanced techniques

such as CT core-flooding and high-frequency pulse-decay have improved efficiency and visualization, but all methods require calibration against reference samples, correction for gas slippage and flow nonlinearity, and cross-checking with alternative test methods or upscaled data. Core-derived permeability reflects the plug scale, typically ranging from centimeters to a meter.

- b. **Log Based Permeability Interpretation:** While wireline logs do not directly measure permeability, log-derived porosity is commonly used as a proxy because porosity logs are continuous, widely available, and far more economical than core analysis. In most clastic reservoirs, engineers rely on the density–neutron overlay for the primary porosity curve, with the sonic log as a secondary check and NMR as a high-confidence reference when available. Additional tools, such as photoelectric factor, dielectric, and capacitive logs, can also provide porosity estimates under certain conditions. However, porosity alone is an incomplete predictor of flow behavior, as it does not account for pore throat geometry, tortuosity, or effective surface area. To address this, permeability is often estimated using empirical relationships that tie porosity to flow capacity, such as the Kozeny-Carman equation (Berryman and Blair 1987; Schulz et al. 2019; Rehman, Hafeez, and Krawczuk 2024).

$$k = \frac{C\phi^3}{(1-\phi)^2\tau^2} \quad (1)$$

In this model, C is a parameter related to particle size, τ is the tortuosity factor, ϕ is porosity, and k is the calculated permeability. Logging tools do not directly measure porosity; rather, they infer it from physical properties such as bulk density (density log), hydrogen index (neutron log), acoustic travel time (sonic log), or fluid response (NMR). Each tool has its own sensitivity to lithology and fluid type, and none can detect tortuosity or effective surface area. As a result, the relationship between porosity and permeability must be empirically established for each rock type, with these geometric and structural effects effectively embedded in the transform coefficients.

The reliability of any log-derived permeability curve depends on two key factors. First, and often the largest source of uncertainty, is the transform model itself. Its coefficients are typically calibrated using a limited number of depth-matched core plug measurements, meaning any bias or inaccuracy in that dataset directly affects the prediction. Second, the resolution and depth of investigation of the derived permeability are constrained by the porosity logs on which the transform is based. Because the transform can only predict permeability within the volume sensed by the porosity tool, any porosity variations outside that depth of investigation remain undetected. Consequently, the output inherits the same vertical scale, smoothing, and limitations as the input logs.

- c. **Permeability derived from NMR logging:** Nuclear Magnetic Resonance (NMR) logging is a non-destructive reservoir characterization technique that offers direct insights into pore structure and fluid distribution. It operates on the behavior of hydrogen nuclei in response to an external magnetic field, making it sensitive to fluids within the pore space. Since hydrogen is abundant in both water and hydrocarbons, NMR tools can quantify fluid-filled porosity without relying on matrix property assumptions. Unlike density and neutron logs, which infer porosity based on rock and fluid properties, NMR measures fluid-filled porosity directly, providing a more accurate representation of the reservoir's storage capacity. NMR porosity is commonly divided into bound fluid porosity and free-fluid porosity, a distinction that is critical for understanding flow potential.

NMR also provides information about pore size distribution through the T_2 relaxation time spectrum (Prammer 1994). Shorter T_2 times indicate smaller pores, while longer T_2 times represent larger pores. This enables detailed characterization of pore throat connectivity and permeability variations. Fig. 1 illustrates the typical T_2 distribution and its subdivisions: clay-bound water (CBW), capillary-bound water (BVI), and free fluid index (FFI). The BVI/FFI cutoff is commonly set to 33 ms in clastics and

100 ms in carbonates, with the actual cutoff calibrated to rock type, fluids, and acquisition conditions (Logan et al. 1998).

Permeability estimation from NMR is achieved using empirical models such as the Timur-Coates (Timur 1968; Timur 1969; Ahmed, Crary, and Coates 1991) and Schlumberger Doll Research (SDR) models (Kenyon et al. 1988; Bryant, Cade, and Mellor 1993; Babadagli 2007). These models relate porosity and T_2 parameters to permeability. The Timur-Coates model estimates permeability from irreducible water saturation and free-fluid porosity, while the SDR model uses the geometric mean of the T_2 distribution and porosity. Both models require calibration constants specific to the formation. MICP (mercury injection capillary pressure) data is often used for comparison, as it reflects pore-throat sizes, whereas NMR reflects pore-body sizes. The Timur-Coates model estimates permeability from free-fluid and bound-fluid porosity:

$$k_{MC} = \left[\frac{\phi}{C} \frac{MFFI}{MBVI} \right] \quad (2)$$

Here, ϕ is the total porosity, MFFI is the free fluid index, MBVI is the bound volume index obtained from CBVI or SBVI methods, and C is a formation-dependent coefficient. The model is flexible and widely used but can over- or underestimate permeability in the presence of hydrocarbons or heavy oil due to mischaracterization of the BVI component. Accordingly, in hydrocarbon-bearing intervals Timur-Coates should only be applied after correcting the T_2 spectrum and porosity for hydrocarbon effects. An HI correction may be required in gas zones, and short T_2 values associated with viscous oil may also cause bias.

The SDR model estimates permeability from the T_2 geometric mean and porosity:

$$k_{SDR} = CT_{2gm}^2 \phi^4 \quad (3)$$

In this case, ϕ is the NMR effective porosity (MPHI), T_{2gm} is the geometric mean of the T_2 distribution, and C is a calibration constant. This model performs well in clean, water-saturated formations. However, in hydrocarbon-bearing zones, the T_2 distribution is skewed, often resulting in permeability overestimation (e.g., with oil filtrates) or underestimation (in unflushed gas zones). Since these hydrocarbon effects are not correctable, the SDR model fails in hydrocarbon-bearing formations.

Both Timur-Coates and SDR estimate matrix permeability and are not suitable for fractured reservoirs. The Timur-Coates model, based on irreducible water saturation and free-fluid porosity, is especially useful in low-permeability settings. The SDR model, relying on T_{2gm} and total porosity, provides a more generalized approach suitable for a broad range of conditions. In shales, where permeability is often below detection limits of conventional logs, NMR-derived estimates help guide well placement and optimize completion strategies (Daigle and Dugan 2009).

Finally, it is important to recognize that NMR has a shallow depth of investigation, typically 0.5 to 1.5 inches, which defines the scale of measured permeability. Advanced tools, such as side-looking wireline NMR, LWD NMR, and slim hydrogeologic probes, may achieve greater depths (Chen and Li 2019; Sun et al. 2019; Delhomme 2014; Kenyon et al. 1995).

- d. **Permeability determination from DST:** Drill Stem Testing (DST) involves deploying a temporary completion string to evaluate reservoir properties before a well is cased or fully completed (Fig. 2). By exposing the formation face without perforation debris or cement sheath, DST provides a direct and representative measurement of permeability, skin factor, initial pressure, and producible fluid types. The test can be conducted in open hole or through casing using a casing tester. Modern DST tool strings incorporate real-time downhole pressure gauges, multiple flow and buildup cycles, large-volume fluid chambers, and surface-controlled annulus pressure, enabling efficient evaluation of heterogeneous intervals. These features allow operators to quickly decide whether to complete, sidetrack, or abandon the tested zone.

Yet the method carries inherent limitations that engineers must keep in mind. First, the flowing period is normally short, frequently only a few hours, which restricts the pressure-transient radius of investigation to tens or hundreds of meters. Any long-radius boundaries, late-time skin changes, or dual-porosity effects simply do not manifest. Extrapolating deliverability beyond the tested drainage volume therefore invites error. Second, wellbore storage is exaggerated. The entire drill string and large annulus volumes must first fill with formation fluids, masking early-time radial flow and making skin and permeability estimates sensitive to tool-volume uncertainty. Third, mud-filtrate invasion and drill solids can severely alter near-wellbore permeability. Even though the borehole is "open," the flow is through an invaded filter cake whose properties are seldom quantified. Fourth, economics and logistics are unfavorable in deep or high-pressure wells. Rig rates are high, and a failed packer or stuck string can rapidly offset any evaluation benefit. Finally, DST data are acquired under an environment of active drilling operations. Mud weight variations, swab-and-surge effects while tripping the string, and rig heave in offshore settings all introduce noise and complicate analysis. Typically, DST investigation radii in clastic reservoirs range from 50 to 500 ft, though deeper penetration is possible depending on test duration and reservoir properties.

- e. **Wireline formation pressure testing (WFPT).** Wireline formation pressure testing (WFT), using tools such as Schlumberger's MDT, Halliburton's RDT, or Baker's RCX, collects brief drawdown/recovery transients at dozens of depth stations while the well is still open hole. Although each "pretest" may last only a few minutes and involve only a few cubic centimeters of flow, the resulting pressure–time trace contains enough information to derive formation pressure, mobility (k/μ), and near-probe skin. Interpreting those data follows the same physics as classical well-test analysis but scaled to centimetric flow domains and dominated by spherical rather than planar radial geometry. Typically, a spherical radial flow is assumed when modeling the pressure change. A characteristic feature of spherical flow is that the pressure-change derivative function shows a $-1/2$ slope on a log–log diagnostic plot. Once the spherical flow regime is identified, the mobility can be calculated using the following relation (Brigharn et al. 1980)

$$\frac{k}{\mu} = \frac{q}{\pi r_w \left(\frac{d\Delta p}{d\sqrt{t}} \right)} \quad (2)$$

where k is the formation permeability, μ is the fluid viscosity, q is the flow rate into the probe, r_w is the radius of the probe, and $\frac{d\Delta p}{d\sqrt{t}}$ is the slope of the pressure change versus the square root of time.

In real scenarios, however, identifying the spherical flow regime is not a trivial task, as the presence of skin factor and wellbore storage effects often distorts it. Moreover, the above model assumes a constant flow rate, while a variable pumping rate can further complicate interpretation and analysis, especially when the test duration is short (Betancourt, Dussan V., and Lake 2010). The case of analysis incorporating both wellbore storage and skin factor has also been well studied (Joseph and Koederitz 1985).

It is very important to understand the radius of investigation using the WFPT tool for the purpose of determining permeability. For spherical flow region, the radius of investigation given by (Schlumberger 2006)

$$r_i = \sqrt{6 \frac{k\Delta t}{\mu\phi c_t}} \quad (3)$$

Where k is determined permeability, Δt is shut-in time; μ is fluid viscosity, ϕ is formation average porosity, c_t is the total compressibility including the pore volume compressibility and average fluid compressibility. All parameters are in oilfield units. Knowing the pretest volume V_p , and considering

the finite resolution of pressure sensor, it is suggested to δ is at about 10 times the gauge resolution and then calculating the Δt_s . If the buildup $\Delta t > \Delta t_s$, the radial radius of investigations is calculated as follows

$$r_{sr} = 0.69 \left(\frac{V_p}{\phi c_t \delta} \sqrt{\frac{k_r}{k_v}} \right)^{1/3} \quad (4)$$

And the vertical radius of investigation is given by

$$r_{sr} = 0.69 \left(\frac{V_p}{\phi c_t \delta} \frac{k_r}{k_v} \right)^{1/3} \quad (5)$$

If the flow regime is radial flow in the test, Δt_s should be obtained on the radial time function, and the depth of investigation for buildup is then given by (Schlumberger 2006)

$$r_r = 0.56 \sqrt{\frac{V_p}{\phi c_t h \delta}} \quad (6)$$

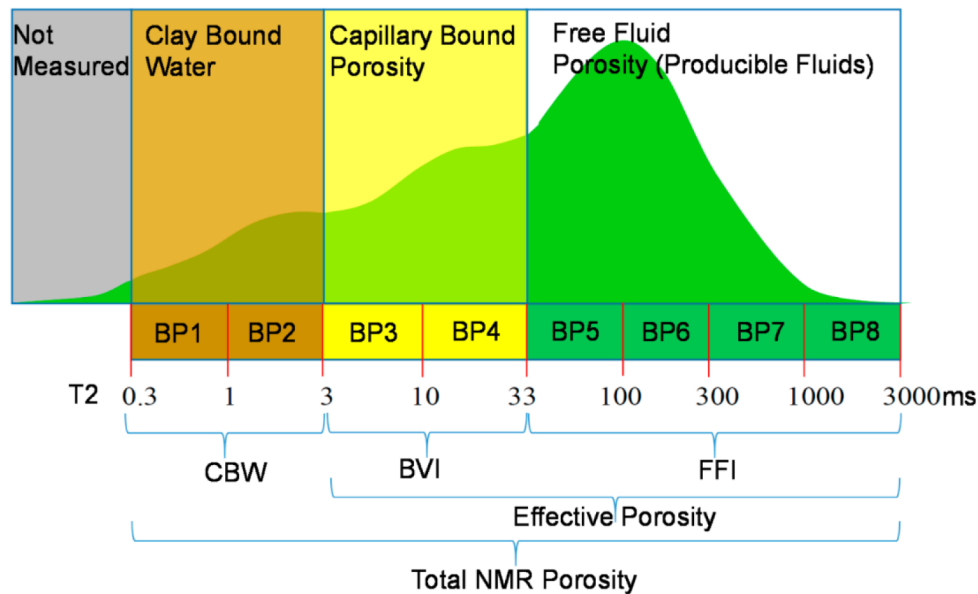


Figure 1—The T2 distribution shows clay-bound water (CBW), capillary bound fluid (BVI), and free fluid (FFI). These properties are shown in relation to the T2 signal (Rezaee 2022).

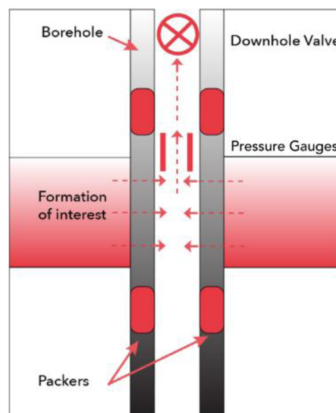


Figure 2—Schematic of a DST tool string showing pressure gauges, test valves, hydraulic bypass or equalizing valves, and packers. The setup isolates a selected interval using packers and allows the formation to flow into the drill string (Source: <https://northstardst.com/services/drill-stem-testing/>, accessed 7/25/2025).

Case Study- Mangaheha-12 Formation and Field

The Mangaheha Formation is a tight gas sandstone reservoir located in the Taranaki Basin of New Zealand. Deposited predominantly in marginal marine to shallow marine and estuarine environments, it comprises interbedded sandstones, carbonaceous mudstones, siltstones, and thin coal layers (Eahsanul Haque et al. 2018; Hossain and Rahman 2025). This fine-scale stratification results in significant vertical heterogeneity, leading to permeability anisotropy driven by lamination and lithological contrasts (Hossain and Rahman 2025). These geological and petrophysical characteristics make the Mangaheha Formation an ideal candidate for evaluating the reliability of wireline formation pressure testing (WFPT) in tight, laminated reservoir settings.

This study focuses on the deviated Mangaheha-12 well, spudded in March 2013 and drilled to a total measured depth (Eahsanul Haque et al. 2018) of 5208 m (4344 m TVDSS). Mangaheha-12 targeted stacked, gas-bearing intervals spanning the MaF to MaA5 subzones of the Mangaheha Formation. The well was completed with a 4½" liner, swell packers, and ten frac sleeves to enable selective hydraulic fracturing and zonal evaluation. Post-completion diagnostics included Production Logging Tool (PLT) surveys and Drill Stem Testing (DST) to assess zonal contributions and pressure response. A comprehensive formation evaluation program was conducted, comprising:

1. **Wireline logs:** Gamma Ray, Density, Neutron, Photoelectric Factor, Resistivity, Acoustic, Nuclear Magnetic Resonance (NMR), resistivity image logs, and production logging.
2. **Wireline Formation Pressure Testing (WFPT):** A Modular Formation Dynamics Tester (MDT) string was deployed in the Mangaheha-12 well to assess in-situ formation pressure, mobility, and fluid type across potential gas-bearing intervals in a tight, low-permeability reservoir. The toolstring configuration featured an Extra-Large Diameter (XLD) probe for improved seal integrity, a high-pressure Pump-Out (MRPO) module for extended fluid extraction, and an AFA (Advanced Fluid Analyzer) unit for downhole compositional fluid analysis. This configuration was specifically optimized for fluid characterization under low-mobility conditions.

Summary of MDT Testing Procedure and Results

The MDT job was performed over a two-day interval, during which 76 volumetric drawdown pretests were attempted in a bottom-up sequence. A breakdown of the test outcomes is presented in Table 1. Each pretest recorded pressure drawdown and recovery curves to estimate formation pressure and mobility. The majority of the tests showed very low mobility (<5 mD/cP), with only a few locations providing valid fluid entry and stable pressure buildup.

Table 1—Summary of MDT pretests in Mangaheha-12 and classification of test quality. Valid tests provided reliable mobility/pressure; tight tests and unstabilized tests failed to stabilize due to low permeability. Dry tests lacked fluid influx despite the seal; no seal tests failed to establish contact. Supercharging refers to artificial elevation of measured pressure due to recent overbalance or mudcake effects, compromising gradient interpretation.

Pretest Status	Count	Description
Valid Tests	22	Clear drawdown and pressure buildup; reliable mobility and formation pressure measured.
Supercharged	5	Pressure buildup trends distorted by wellbore pressure influence; pressure appears artificially elevated.
Tight Tests	25	Formation response too slow; mobility too low for pressure to stabilize within test duration.
Dry Tests	8	Probe sealed but no fluid influx observed; typically indicates gas or highly resistive dry zone.
No Seal	9	The probe failed to form a pressure-tight seal against the borehole wall—no valid test possible.
Unstabilized	7	Drawdown occurred but buildup did not stabilize within the timeframe; data considered inconclusive.
Total	76	—

The key technical findings are summarized below:

1. **Low Formation Permeability:** Most intervals were classified as tight. The maximum pretest mobility recorded was ~ 5.2 mD/cP at 4416.5 m MD. However, post-DFA cleanup mobility at the same depth reached 163.3 mD/cP, indicating significant fluid contamination prior to cleanup.
2. **Supercharging Effects:** Approximately 42 of 76 pretests showed signs of supercharging, which led to poor pressure buildup quality and no reliable pore pressure gradient could be derived.
3. **DFA Station Outcomes:** Four pump-out DFA stations were conducted:
 - a. 4824.5 m MD: Identified gas presence with increasing GOR and reducing C6+ fraction. Pretest mobility was 3.54 mD/cP; post-cleanup mobility improved to 65.4 mD/cP.
 - b. 4804.0 m MD: Aborted after 40 minutes due to insufficient cleanup at a mobility of ~ 1.78 mD/cP.
 - c. 4495.3 m MD: Gradual water entry observed, suggesting transition to water-bearing zone. Pump rate reduced due to increased drawdown from 700 psi to 2400 psi.
 - d. 4418.0 m MD: Confirmed gas presence with excellent cleanup. Post-DFA mobility reached 163.3 mD/cP, the highest recorded.
4. **Fluid Interpretation:** The AFA data indicated that gas was the dominant component in at least two DFA stations. The decline in C6+ and appearance of formation water in one test suggest fluid gradients and possible compositional grading.
3. **Coring:** 48 sidewall core samples recovered.
4. **Well testing:** Drill Stem Test (DST) across multiple frac stages.

To benchmark WFPT-derived permeability and evaluate scale-dependent differences, permeability estimates were compared across six reservoir intervals: MaD1-A, MaD1-B, MaC3, MaB1, MaA5u, and MaA5l using the following methods:

1. Core Plug Permeability (Matrix-Scale Ground Truth)

Permeability at reservoir conditions was measured from core plug samples in select intervals and served as the reference for validating other methods. These measurements reflect matrix-scale permeability unaffected by fluid mobility effects or near-wellbore disturbances. However, due to sparse core coverage, the results may under-represent contributions from fractures or thin, high-permeability laminations.

2. Log-Based Permeability Estimates

- o **Porosity–Permeability Transform (Poro-Perm):** Permeability was estimated from neutron and density porosity logs using a transform calibrated to local core data. This approach performs well in clean sandstone intervals but tends to overestimate flow capacity in laminated or capillary-bound zones due to its limited sensitivity to flow barriers.
- o **Nuclear Magnetic Resonance (NMR):** Permeability was also estimated using the Timur-Coates model, which relates permeability to the T_2 distribution, bound fluid volume (BVI), and free fluid index (FFI). Field-default constants were used. While NMR logging provided continuous permeability profiles across multiple zones, the model assumes uniform wetting and connectivity, assumptions that often break down in laminated formations.

3. Wireline formation pressure testing (WFPT)

In this study, 76 Modular Formation Dynamics Tester (MDT) pretests were performed between the measured depths of 4178 m and 5117.5 m using a wireline-deployed tool with a single-probe module and dual pump chambers. Depths were selected to sample a range of laminated and non-laminated lithofacies based on image logs and porosity. Each test included: (1) probe sealing and contact check,

(2) low-rate drawdown (2–6 cc/s) until pressure stabilized or drawdown exceeded 400 psi, and (3) pressure buildup until flattening or telemetry timeout.

Tests were classified into six categories: valid, tight, supercharged, dry, lost seal, or unstabilized. This classification is consistent with established guidelines, where "valid" tests exhibit stable drawdown and buildup with clear pressure stabilization as can be seen in Figure A1.1 with its parameters in Table A1.1, "tight" tests show slow or insufficient pressure responses indicative of ultra-low permeability as can be seen in Figure A1.2 with its parameters in Table A1.2, and "supercharged" tests present pressures elevated above true formation pressures due to mud filtrate invasion effects as can be seen in Figure A1.3 with its parameters in Table A1.3 (Proett et al. 2014). Six high-quality ("valid") tests, demonstrating stable drawdown curves and consistent pressure stabilization, were selected for detailed analysis, providing reliable data critical for interpreting reservoir characteristics and WFPT efficacy in tight, laminated formations. Drawdown and buildup data from these were interpreted using two approaches.

- **Analytical approach (spherical flow equation):** Permeability was first estimated analytically using the spherical flow equation under the assumption of early-time, homogeneous, isotropic flow conditions. This served as a baseline approximation, although such estimates are sensitive to probe storage, near-wellbore effects, and flow geometry.

$$k_{dd} = C_{pf} \frac{q\mu}{\Delta p} \quad (9)$$

Where q is the drawdown rate (bbl/day), μ is fluid viscosity (cP), and Δp is drawdown pressure (psi). The shape factor C_{pf} reflects probe geometry and flow configuration and is critical to translating drawdown behavior into permeability estimates. (Weinheber et al. 2008) reported empirical C_{pf} values ranging from 856 for an elliptical probe (1.45 in equivalent radius) to 5660 for a conventional 0.22 in probe. These values are derived under standard test assumptions and incorporate unit scaling, providing a robust basis for analytical permeability estimation across different tool configurations.

- **Numerical Modeling:** To capture directional permeabilities (including radial permeability and vertical permeability) and determine flow geometry, a numerical inversion approach was applied using a composite radial–axial flow model. This model simulates pressure response by transitioning from spherical to radial flow, while accounting for skin and storage effects. A limited-entry condition was imposed vertically to constrain the vertical-to-horizontal permeability ratio (k_z/k_r), consistent with image log observations of laminated barriers.

Key model inputs are presented in Table A1.4 for each selected pretest. Permeability and anisotropy were derived by adjusting k_r , k_z/k_r , skin, and hydraulic window thickness (Motaei et al. 2023) to fit observed pressure and derivative data. The system was modeled as infinite-acting laterally with vertical no-flow boundaries.

Due to the short duration of many tests (<5 minutes), late-time radial flow was often not reached, increasing uncertainty in k_r and k_z/k_r . In these cases, k_z/k_r was used as a qualitative indicator of connectivity. Multiple parameter sets (e.g., high anisotropy with small H_w) could produce similar matches, reflecting non-uniqueness in inversion. Despite this, consistent differences in directional flow behavior emerged, with laminated intervals such as MaB1 and MaA5u requiring k_z/k_r values >150 for acceptable matches.

4. DST-derived permeability

Where available, Drill Stem Testing (DST) was used to estimate permeability at the zone level across hydraulically stimulated intervals. These tests reflect post-fracture deliverability under dynamic flowing conditions and integrate contributions from multiple sublayers. However,

due to their large radius of investigation and the composite nature of stimulated zones, DST-derived permeability can obscure small-scale heterogeneities captured by core, NMR, or WFPT measurements.

Uncertainty and Anisotropy Interpretation in WFPT Analysis

The interpretation of permeability and anisotropy from wireline formation pressure testing (WFPT) is subject to several sources of uncertainty, especially in low-permeability, laminated formations such as the Mangahewa. Key sources of uncertainty include:

- **Seal integrity and tool standoff:** In laminated intervals, even slight misalignment can lead to failed seals or masked responses.
- **Supercharging:** Occurred in several zones due to previous overpumping or poor mudcake development, leading to overestimated formation pressure and distorted drawdown/buildup profiles.
- **Storage and skin effects:** Short test durations in tight formations often limit the ability to fully capture late-time radial flow, resulting in ambiguous permeability fits.
- **Flow regime misclassification:** Many tests appear dominated by spherical flow or limited-entry behavior, but lamination may suppress vertical flow, leading to high k_z/k_r estimates that are hard to validate independently.

Despite these challenges, numerical inversion modeling provided useful qualitative indicators of anisotropy. For example, in MaB1 and MaA5u, pressure transients could only be matched when vertical-to-horizontal permeability ratios (k_z/k_r) exceeded 150. These high values are consistent with thinly laminated siltstone barriers and are supported by borehole image logs showing minimal vertical conductive pathways.

An additional source of uncertainty stems from the differences in depth of investigation across measurement methods. WFPT captures pressure behavior from the immediate near-wellbore region, typically within a few inches, making it highly sensitive to localized heterogeneity, tool standoff, and mudcake effects. In contrast, NMR logs average properties over a broader interval, often several feet, while DSTs reflect the flow behavior of entire completed zones. This mismatch in investigation scale can lead to discrepancies in interpreted permeability, especially in laminated formations where thin high-permeability streaks or vertical flow barriers may be sampled inconsistently. Understanding these differences is essential when reconciling WFPT results with other data sources.

Because of test limitations, k_z/k_r and H_w values are interpreted as qualitative constraints rather than definitive parameters. Their consistency with DST production rankings (for example, MaD1 outperforming MaB1) supports their use as relative indicators of connectivity. This highlights the need to integrate WFPT interpretation with core, NMR, and production data to reduce ambiguity in low-permeability, anisotropic formations.

Results and Discussion

We compared permeability estimates and assessed vertical communication indicators across six reservoir intervals (MaD1-A, MaD1-B, MaC3, MaB1, MaA5u, and MaA5l). Permeability estimates were derived from logs (NMR and porosity-permeability transforms), core plugs, Drill Stem Tests (DST), and Wireline Formation Pressure Testing (WFPT), using both spherical flow equations and pressure derivative analysis via numerical modeling.

Permeability comparison across methods

Static methods (core plugs, NMR, and porosity–permeability transforms) produced internally consistent permeability estimates, particularly in the moderate-quality MaD1-B interval, where values clustered between 100–112 mD. In contrast, dynamic measurements yielded significantly lower values: ~60 mD from MDT spherical-flow interpretation and only 6.1 mD from DST.

In lower-permeability zones such as MaB1 and MaA5u, all methods indicated tight rock conditions, with core and poro–perm estimates below 6 mD. MDT-derived permeabilities were especially suppressed, ranging from 0.01 to 0.08 mD, primarily due to near-wellbore effects and poor pressure signal propagation in tight rock. This discrepancy reflects the nature of MDT pretest measurements, where drawdown mobility (k/μ) is recorded under conditions highly sensitive to mud filtrate invasion and fines migration. These near-probe disturbances often result in underestimated apparent permeability, particularly in low-mobility formations where even minor damage impairs flow response. In contrast, core analyses are typically performed on cleaned, undamaged samples, yielding higher values more representative of intrinsic matrix permeability (Dussan V. and Sharma 1992; [Betancourt, Dussan V., and Lake 2010](#)).

Accurate spherical flow interpretation also depends on tool design and test configuration. Larger probe radii and optimized flow volumes improve the chances of capturing a measurable signal in formations with mobility between 0.01 and 10 mD/cP. However, in ultra-tight rocks (<0.01 mD/cP), spherical flow is rarely sustained, and the signal remains dominated by near-probe distortions. This challenge is compounded in environments with significant filtrate contamination, where the pressure response can be skewed ([Seifert et al. 2005](#); [Betancourt, Dussan V., and Lake 2010](#)).

Moreover, the different depths of investigation among methods contribute to the divergence. Core and NMR data represent centimeter-scale volumes, while MDT probes millimeter-scale flow near the borehole, and DST integrates over tens of meters. As such, DST often averages out heterogeneities, dampening the influence of localized high-permeability streaks detectable by near-wellbore methods.

These findings emphasize that differences between static and dynamic measurements do not imply inconsistency but rather reflect the interplay between flow geometry, damage scale, and data resolution. Proper interpretation requires an integrated understanding of test scale, formation heterogeneity, and mobility-dependent behavior.

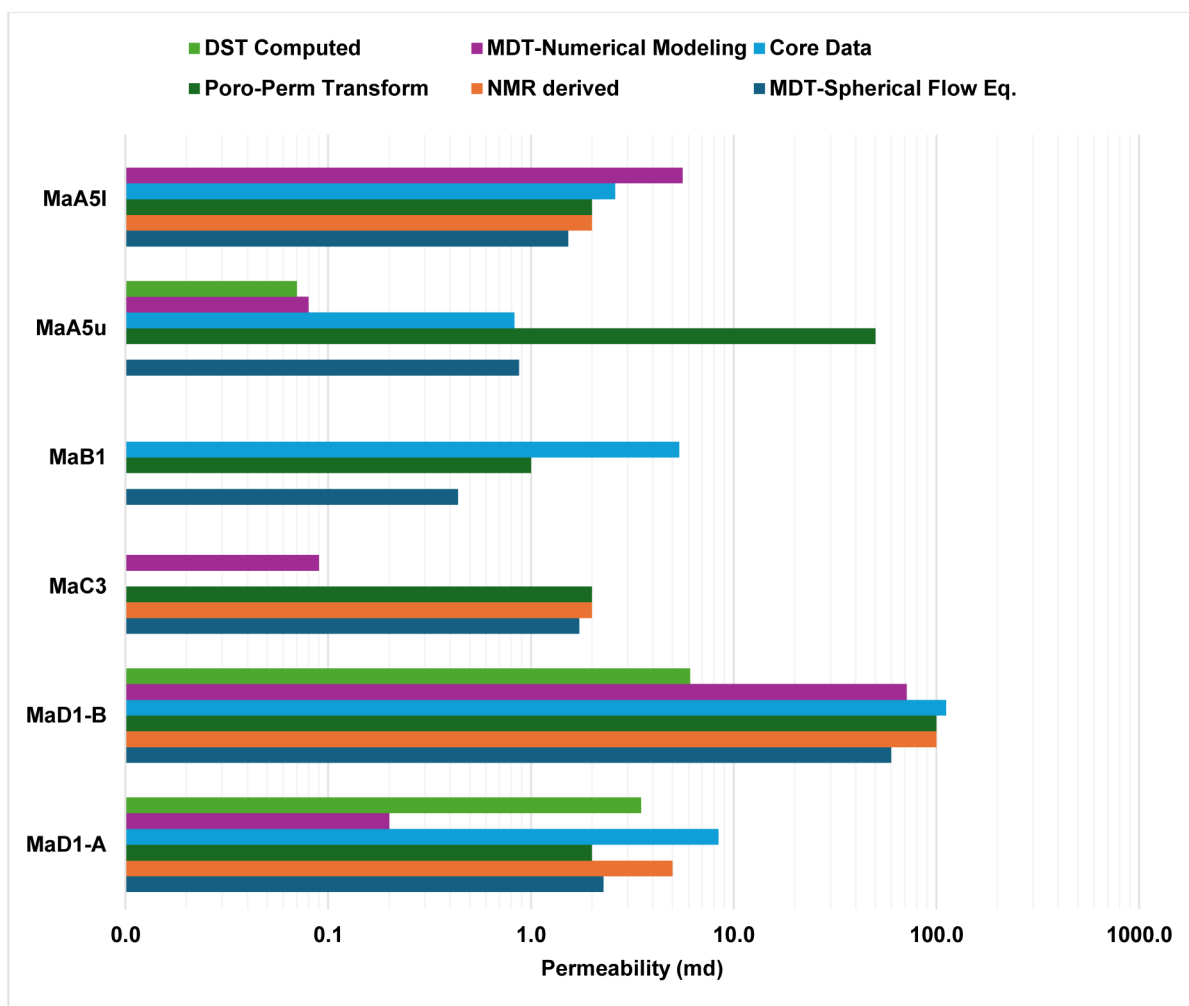


Figure 3—Permeability values for each zone using all six methods, emphasizing the discrepancies between core/lab-based and pressure-transient-derived estimates.

Vertical permeability ratio and hydraulic window

Vertical permeability ratios (k_z/k_r) and hydraulic window thickness (Motaei et al. 2023), extracted from numerical modeling using a composite radial-axial flow approach, provide insights into vertical connectivity and anisotropy. In zones such as MaB1 and MaA5u, extremely high k_z/k_r values (348 and 160, respectively) were observed; however, these values can be misleading when absolute permeability is very low. In such cases, the high ratio reflects strong anisotropic resistance rather than meaningful vertical flow.

In contrast, MaD1-B exhibited a k_z/k_r ratio of 0.01 combined with high lateral permeability, consistent with a well-laminated, laterally extensive reservoir with limited vertical transmissibility. These contrasting results emphasize that k_z/k_r and H_w are best interpreted as qualitative indicators of flow geometry, not direct measures of vertical transmissibility.

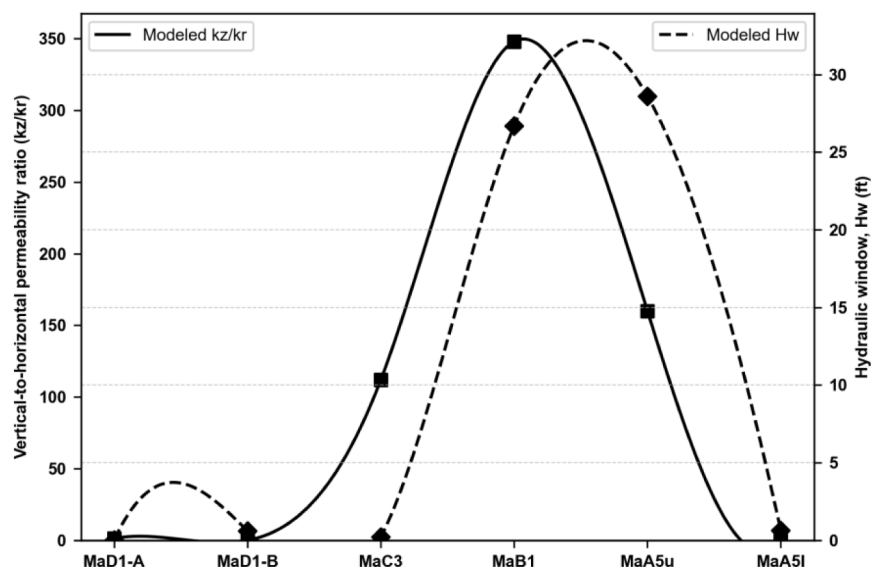


Figure 4—Vertical-to-horizontal permeability ratio (k_z/k_r) and hydraulic window (Motaei et al.) for each analyzed zone, derived from numerical modeling. Illustrates the need for geological context when interpreting anisotropy.

Flow regime behavior from pressure derivative:

Pressure and derivative responses from MDT pretests were analyzed to identify dominant flow regimes. In the MaD1-A interval (4416 m), the derivative curve exhibits an early-time slope near $-1/2$ suggestive of linear or limited-entry flow, followed by a flattening trend that may indicate a transition toward radial or pseudo-radial behavior. While these features align with theoretical expectations, the short duration of the test, the presence of noise, and laminated formation characteristics introduce significant uncertainty.

As a result, although the derivative trends point to possible anisotropic flow transitioning toward radial geometry, this interpretation remains tentative and should be corroborated by longer-duration testing or supporting datasets.

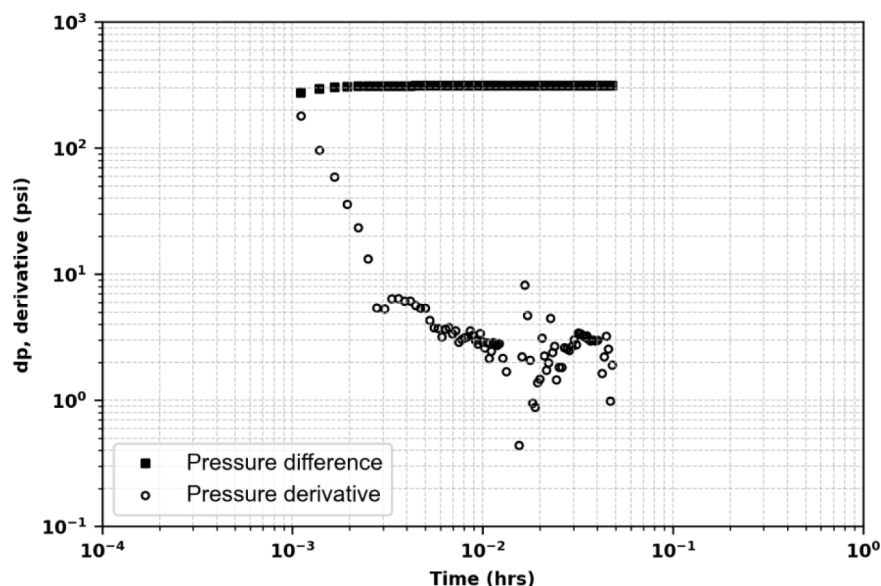


Figure 5—MDT pressure and derivative log-log plot from MaD1-A (4416 m). Slope transition may indicate anisotropic-to-radial flow behavior, though interpretation is limited by short test duration.

Conclusion

This study systematically evaluates the MDT-derived permeability in tight, laminated formations across six reservoir zones in the Mangahewa-12 well, and compares the values with static and other dynamic methods. While drawdown-derived permeability shows reasonable agreement with NMR estimates in moderate-quality intervals, interpretation reliability declines in low-permeability, anisotropic zones due to vertical heterogeneity, short test durations, and ambiguous flow regimes.

Numerical-inversion outputs, such as the vertical-to-horizontal permeability ratio and the hydraulic-window thickness, illuminate directional flow potential, yet their diagnostic power is bounded by the very small volume they sample (typically centimeters around the probe). Log-based porosity-permeability transforms operate on the same thin scale and therefore exaggerate permeability when bound fluids or disconnected vugs dominate the investigated pores. Drill-stem tests (DSTs), by contrast, interrogate a drainage radius that can extend tens of meters, integrating multiple beds and completion effects. This large-scale averaging often masks high-permeability streaks and makes interval-specific deliverability appear lower than suggested by core or log data. In short, the apparent inconsistencies among numerical models, log transforms, and DST results are driven less by tool accuracy than by the radically different investigation scales each method addresses.

The same scale issue governs the reliability of formation tester measurements in laminated reservoirs. WFPT probes only a few decimetres into the formation, so its permeability estimates are meaningful only when calibrated against core plugs and logging data and interpreted within a geologic framework that recognises bed-scale heterogeneity. By integrating small-scale (core, logs, MDT) and large-scale (DST) observations, operators can place WFPT data into reservoir models with the right level of confidence—avoiding the pitfalls of over-interpreting isolated pressure transients and enabling more robust development decisions in stratified gas systems where data remain sparse.

ACKNOWLEDGEMENT

In analyzing the data, Kappa software licensed to the University of Oklahoma was used to support the case study.

Nomenclature

C	= Particle size
C_{pf}	= Shape factor
c_t	= Total compressibility (1/psi)
Δt	= Shut-in time (minutes)
k	= Permeability (mD)
k_z	= Vertical permeability (mD)
k_r	= Radial permeability (mD)
μ	= Fluid viscosity (cP)
ϕ	= Porosity
q	= Flow rate (bbl/day)
r_w	= Wellbore radius
T_2	= Relaxation time spectrum (milliseconds)
T_{2gm}	= Geometric mean of the T_2 distribution (milliseconds)
τ	= Tortuosity
H_w	= Hydraulic window (ft)

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Appendix

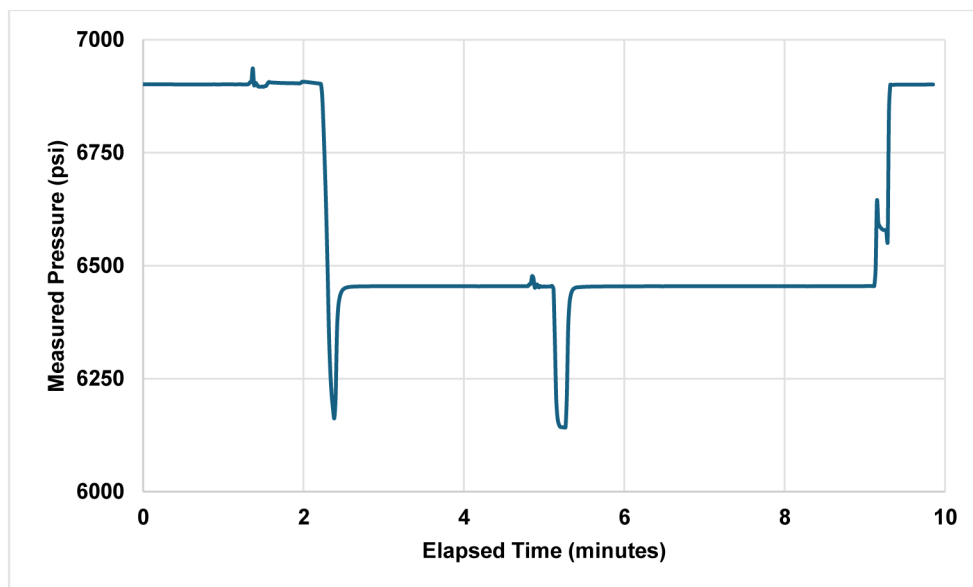


Figure A1.1—Pressure response during formation test at 4416.5 m MD (MAD1-A). Drawdown and buildup behavior recorded with an XLarge-Diameter probe, stabilizing at 6454.97 psi.

Table A1.1—Test parameters at 4416.5 m MD. Includes probe type, pretest conditions, formation pressure, temperature, and test validity.

Probe Type	Pretest Type	Formation Pressure (psi)	Pretest Rate (bbl/day)	Pumped volume (cc)	Temp (Deg. C)	Test Type
XLarge-Diameter	Drawdown	6454.97	0.54	9.59	124.4	Valid Test

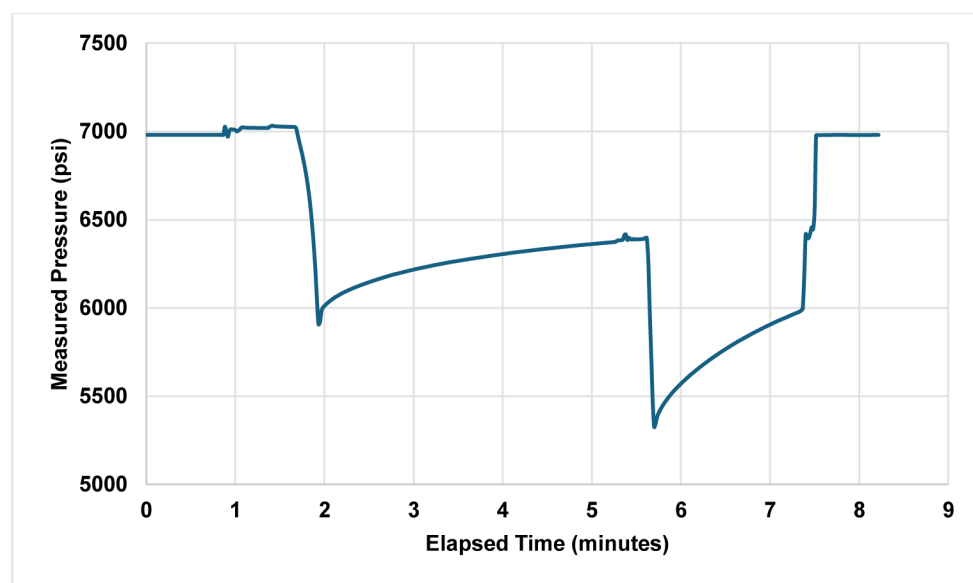


Figure A1.2—Pressure response during formation test at 4461.0 m MD (MAD1-A). Drawdown and attempted buildup behavior recorded with an XLarge-Diameter probe. No stable pressure achieved; test classified as tight.

Table A1.2—Test parameters at 4461.0 m MD. Includes probe type, pretest conditions, pumped volume, temperature, and test validity. Lack of pressure stabilization indicates a tight formation.

Probe Type	Pretest Type	Formation Pressure (psi)	Pretest Rate (bbl/day)	Pumped volume (cc)	Temp (Deg. C)	Test Type
XLarge-Diameter	Drawdown	N/A	0.21	1.94	125.5	Tight Test

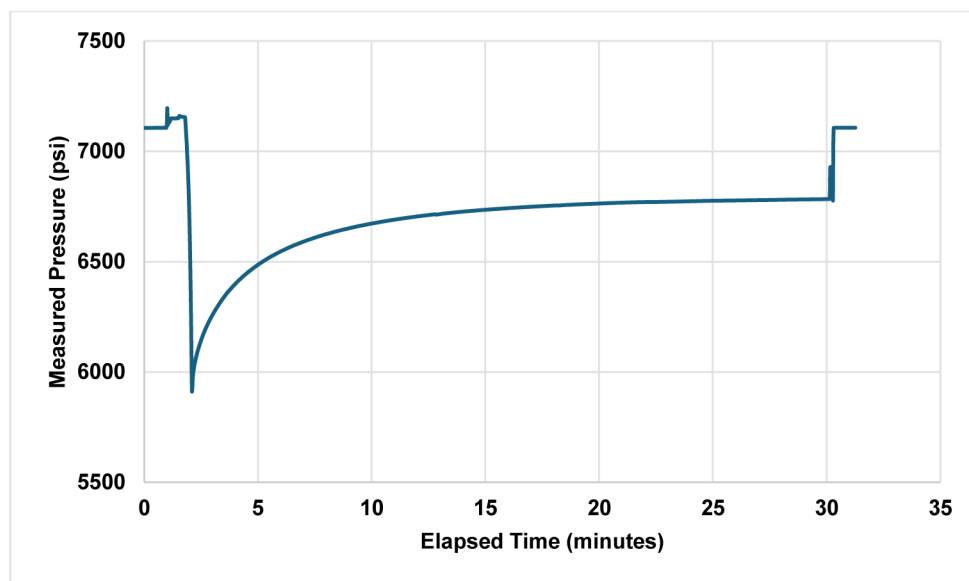


Figure A1.3—Pressure response during formation test at 4532.03 m MD. Drawdown and extended buildup behavior recorded with an XLarge-Diameter probe. Pressure did not stabilize within the test duration, indicating possible supercharging.

Table A1.3—Test summary at 4532.03 m MD. Includes probe type, pretest conditions, pumped volume, temperature, and test validity. Pressure trend suggests a *Supercharged* test response.

Probe Type	Pretest Type	Formation Pressure (psi)	Pretest Rate (bbl/day)	Pumped volume (cc)	Temp (Deg. C)	Test Type
XLarge-Diameter	Drawdown	N/A	0.22	9.32	126.4	Supercharged

Table A1.4—Model input parameters for numerical analysis. Includes interval-specific porosity and pay thickness values. Well radius, rock compressibility, and fluid viscosity were held constant across all intervals.

Interval	Well Radius (in)	Rock Compressibility (psi^{-1})	Fluid Viscosity (cP)	Porosity (fraction)	Pay Thickness (ft)
MaD1-A	6.125	3.0×10^{-6}	0.22	0.08	16.4
MaD1-B				0.13	2.5
MaC3				0.08	32.8
MaB1				0.06	65.6
MaA5u				0.08	66.2
MaA5l				0.07	9.8